

ILYA CHEKHOV

Lead Python Backend Engineer / Tech Lead

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Open to fully remote roles worldwide

Professional Summary

Lead Python Backend Engineer / Tech Lead with 6+ years of commercial experience designing and scaling high-load, event-driven backend systems and microservice architectures. Strong hands-on expertise in Python, FastAPI, Django, PostgreSQL, RabbitMQ, Redis, Kafka, Kubernetes, and distributed systems, with end-to-end ownership spanning architecture design, production engineering, service integrations, reliability, and performance optimization. Experienced in establishing engineering standards, technical specifications, code review, mentoring, and cross-service architecture. Delivered systems serving 630,000+ users and measurable improvements including up to 60% lower database load, 50% lower RabbitMQ memory consumption, and 20% fewer message-processing failures.

Technical Skills

Languages & Frameworks: Python, FastAPI, Django, Django REST Framework (DRF), asyncio, SQLAlchemy, Alembic, Celery

Data & Messaging: PostgreSQL, ClickHouse, Redis, RabbitMQ, Apache Kafka, gRPC

Infrastructure & DevOps: Docker, Kubernetes (K8s), Helm, Ansible, GitLab CI/CD, Linux

Observability & Auth: Grafana, Prometheus, Keycloak, Vault

Architecture & Practices: Microservices architecture, REST API design, system design, distributed systems, Pytest, LLM integration

Engineering Leadership: Technical leadership, architecture design, code review, mentoring, onboarding, technical specifications, production incident management

Work Experience

Lead Python Developer — X5 Group

Remote | Jun 2025 – Present

Food-media and e-commerce platform of Russia's largest food retailer; microservices backend of approximately 20 Python/FastAPI services.

- Designed and built a video-processing service end to end, implementing chunked uploads to S3-compatible storage, asynchronous FFmpeg transcoding into four renditions (1080p–360p), processing-state management, and HLS adaptive streaming.
- Designed and implemented a content recommendation feed integrating X5's centralized recommendation service through Keycloak-authenticated service-to-service communication with token caching; added an in-memory TTL-cached cold-start strategy, eliminating per-request database queries for new users.
- Introduced RabbitMQ as the platform-wide event-driven messaging layer from scratch, establishing team-wide producer/consumer standards with FastStream, routing-key conventions, dead-letter handling, CI testing, and declarative exchange/queue topology managed through Helm.
- Designed and deployed a production RabbitMQ cluster on Kubernetes using the RabbitMQ Cluster Operator, 3 replicas, quorum queues, Vault-managed credentials, and Prometheus monitoring.
- Built an idempotent master-data synchronization pipeline for product-category hierarchies across three retail chains, using batched PostgreSQL upserts, two-pass parent-hierarchy resolution, and transactional rollback on failure.
- Accelerated test execution and CI using pytest-xdist with loadscope distribution and incremental test selection via pytest-testmon, with enforced coverage gates in GitLab CI.
- Managed Helm-based Kubernetes configuration for approximately 20 microservices and 26 CronJobs supporting product imports, data synchronization, and database maintenance across production and 18 staging environments.

Lead Python Developer / Team Lead — Skillbox

Remote | Dec 2022 – May 2025

One of Russia's largest EdTech platforms (#1 in continuing-professional-education training quality, Smart Ranking); 630,000+ users.

- Led architecture design, technical specifications, release management, and production monitoring for the core learning platform and its supporting microservices serving 630,000+ users.
- Optimized PostgreSQL performance through query, schema, and indexing improvements, cutting database load by up to 60%.
- Reduced RabbitMQ memory consumption by up to 50% and message-worker failures by 20% through queue architecture and consumer tuning.
- Delivered CRM (S2) and CDP (Mindbox) integrations and migrated transactional and marketing email delivery to a new SMTP provider, unifying customer data and communications across the platform.
- Refactored the core lessons entity from a foreign-key to a many-to-many data model, improving data-model flexibility and query performance.
- Mentored and onboarded engineers; established code-review standards and engineering best practices across the team.

Senior Python Developer — Playrix

Remote | Jul 2021 – Dec 2022

Global mobile-gaming company (Gardenscapes, Homescapes).

- Built and maintained internal enterprise services: an employee self-service portal, electronic document management (EDM) integration, and a game-data management platform.
- Automated internal business processes, reducing manual operational workload across departments.

Python Developer — Intekh

Remote | May 2021 – Jul 2021

- Developed backend microservices for a self-service warehouse platform: mobile-app smart-lock access, automated payment deduction, and inventory tracking.

Python Developer — Tensor

Yaroslavl, Russia | Dec 2019 – May 2021

Enterprise software vendor (accounting and reporting systems).

- Developed and maintained a management-accounting system and automated tax-reporting workflows for enterprise clients.

Education

Specialist Degree (5-year university degree), Computer Security

2020

Saint Petersburg Electrotechnical University “LETI”, Saint Petersburg, Russia

Languages

Russian — native | **English** — professional working proficiency (CEFR B2)